



UyirNadi

PROJECT GUIDE · HEALTHCARE MANAGEMENT SYSTEM

Your Lifeline to Better Health.

A plain-language guide to the UyirNadi web application: what it does, who uses it, how the screens work, and how health information is safely connected to Firebase and AI services.

நோய்நாடி நோய்முதல் நாடி அதுதணிக்கும் வாய்நாடி வாய்ப்பச் செயல்.

Audience: product owners, healthcare teams, administrators, and non-technical stakeholders

Application: Next.js web application with Firebase services and optional Gemini AI

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1. What UyirNadi is

UyirNadi is a responsive healthcare-management website. It gives a person one place to track everyday health information, prepare for medical appointments, keep documents private, and get clearly labelled informational AI guidance. It also gives doctors and administrators their own workspaces.

The idea in simple terms: a patient sees their own health dashboard; a doctor sees only patients connected through approved appointments; an administrator manages people, appointments, education content, and emergency contacts.

What the product does not do

- It does not diagnose illness, prescribe medication, or change a dosage.
- It does not contact emergency services automatically.
- It does not present AI output as a medical decision.
- It does not give doctors access to private reports until the patient shares them and an approved appointment relationship exists.

Who uses it

Patient / user

Tracks wellbeing, stores documents, books appointments, reads health education, and uses the AI tools as informational support.

Doctor

Manages availability and appointments, sees relevant connected patients, and views only reports the patient has deliberately shared.

Administrator

Creates accounts, manages doctor profiles and appointments, publishes health tips, manages feedback and emergency contacts, and sees platform-level metrics.

Platform owner

Configures Firebase, storage, security rules, the optional Gemini key, and production deployment settings.

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2. Technology used

UyirNadi is one web application. There is no separate Express server to maintain. The website and its secure server routes are built together using Next.js.

Technology	What it is used for	Why it matters to a non-technical owner
Next.js 16 with TypeScript	The website, pages, protected server routes, and production build.	One maintainable application for both the visible website and secure business actions.
React	Interactive forms, dashboards, charts, upload flows, and mobile navigation.	Pages update smoothly without full-page reloads.
Tailwind CSS and reusable UI components	Consistent spacing, colours, cards, buttons, forms, dialogs, empty states, and loading states.	Creates a consistent premium healthcare appearance on phone, tablet, and desktop.
Firebase Authentication	Sign in, registration, password reset, persistent sessions, and role claims.	Accounts are handled by Google's authentication service rather than stored as passwords in the app database.
Cloud Firestore	Private health records, appointments, profiles, health tips, feedback, and notifications.	A cloud database that updates the app with current data.
Firebase Storage	Private report, prescription, and profile-image files.	Separates large private files from normal database records. It must be enabled in Firebase before uploads are live.
Firebase Admin SDK	Secure server-only account creation, sessions, role checks, and protected data actions.	The browser never receives an administrator key.
Gemini API (optional configuration)	Health chat, symptom guidance, nutrition plans, report explanations, and prescription extraction.	AI runs on the server only and is unavailable until a Gemini key is configured.
OCR provider interface	Reads text from uploaded PDF/JPEG/PNG files; Gemini Vision is the initial provider.	The OCR supplier can be changed later without rebuilding the upload screens.
Recharts	Water, weight, fitness, and health-trend charts.	Turns saved information into simple visual progress views.

Important configuration status: Firebase Authentication and Firestore are connected. Cloud Storage must be enabled in Firebase for real file uploads. A server-only `GEMINI_API_KEY` must be added before live AI responses, OCR, and report analysis can run.

3. The user interface

The visual design uses light backgrounds, dark readable text, teal and green healthcare accents, generous spacing, subtle borders, and the supplied UyirNadi logo. The design adapts intentionally for each screen size.

How navigation works

Desktop and laptop A fixed sidebar gives quick access to the dashboard, health tools, appointments, documents, education, profile, and feedback. The top area contains search and notifications.	Phone A bottom navigation shows the main destinations. A collapsible menu keeps less-frequent features available without crowding a small screen.
Loading and empty views Skeleton placeholders appear during loading. Empty sections explain what is missing and provide a direct action, such as “Upload report” or “Book appointment”.	Accessibility Forms have labels and error messages, controls work with a keyboard, and colour is not the only way information is communicated.

Main screens for a patient

Screen	What the person sees and can do
Dashboard	A personal greeting, health summary, BMI, water progress, fitness summary, recent reports, upcoming appointment, health tips, and quick actions.
Health analytics	Charts for water intake, activity, weight/BMI history, recent documents, and appointment trends using saved data.
AI assistant	A ChatGPT-style conversation view with message history, timestamps, a new-conversation action, safety wording, and a typing state.
Prescription reader and reports	Secure upload, preview, consent for AI analysis, editable extraction results, clearly stated uncertainty, and saved document history.
Appointments	Find a doctor, view available dates and slots, confirm an appointment, and review appointment status.
Daily wellbeing	BMI calculator, water tracker, fitness log, nutrition plans, emergency contacts, health tips, profile, and feedback.

Staff screens

Doctor: dashboard, appointments, availability, patients, shared reports, prescriptions, and profile.

Administrator: dashboard, users, doctors, appointments, health tips, feedback, emergency contacts, and

analytics.

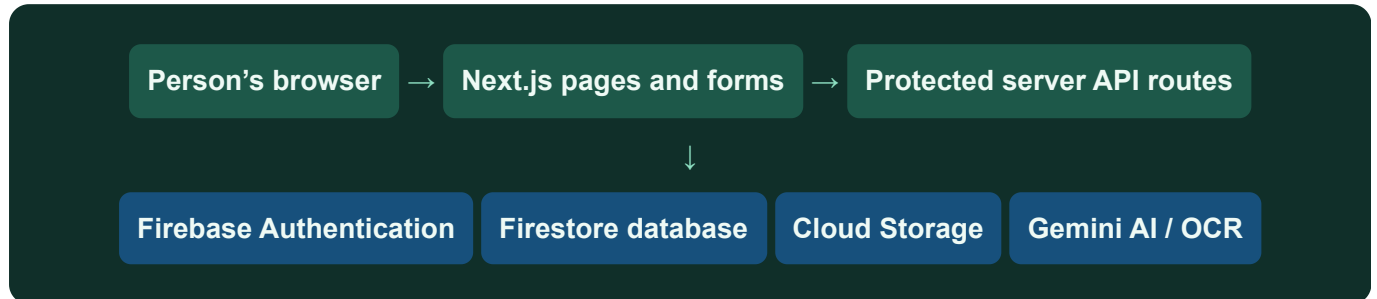
4. Feature guide

Feature	What happens	Safety or privacy behaviour
Registration and sign-in	A person can register, sign in, sign out, and request a password reset. Google sign-in is available when configured in Firebase.	Firebase manages authentication. The app creates a secure httpOnly session cookie after sign-in.
Profile	The user can save basic contact and wellness details such as height, weight, dietary preference, and emergency contact.	Form validation checks the submitted information before it is saved.
BMI, water, and fitness	The user enters height/weight, adds water, and logs activity. The dashboard and charts use the saved history.	BMI is described as a general screening measure. Calories are not presented as medically exact.
Health reports	PDF, JPG, JPEG, and PNG files can be uploaded, privately stored, and optionally analysed after consent.	File type, file signature, size (maximum 8 MB), ownership path, and sharing rules are checked. Reference ranges are never invented.
Prescription reader	The system extracts medicine name, dosage, frequency, duration, and instructions. The user can edit and confirm the result.	The result is labelled AI-extracted, may include low-confidence fields, and is never claimed to be medically verified.
Symptom checker	The user supplies symptoms, severity, duration, and optional context. The AI returns possible explanations and an urgency category.	It never gives a confirmed diagnosis. Emergency patterns receive prominent advice to seek immediate professional care.
Nutrition planner	A plan can be generated from food preferences, goals, activity, allergies, and optional health context.	It is general information and reminds users to seek qualified nutrition advice when relevant.
Appointments	Users choose an available doctor slot. Doctors manage availability and appointment status.	A Firestore transaction re-checks the slot immediately before booking to prevent double booking.
Report sharing	A patient chooses which approved doctor can see a report.	The doctor needs both a confirmed/completed appointment relationship and an explicit share entry.
Tips, feedback, emergency contacts	Users read published educational content, submit feedback, and view administrator-managed emergency information.	The emergency page does not claim that an emergency call has been placed.

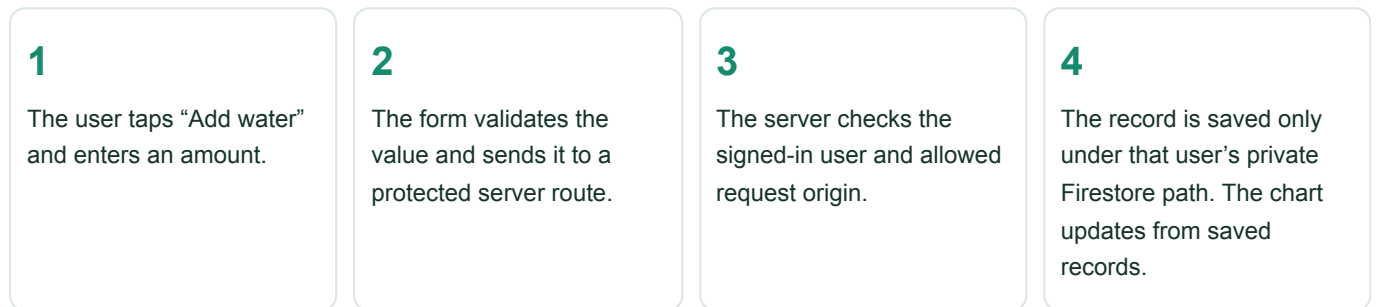
Removed from this version: the medicine-reminder feature was intentionally removed at the product owner's request. The prescription workspace remains available for document review and safe storage.

5. How the website connects to the backend and database

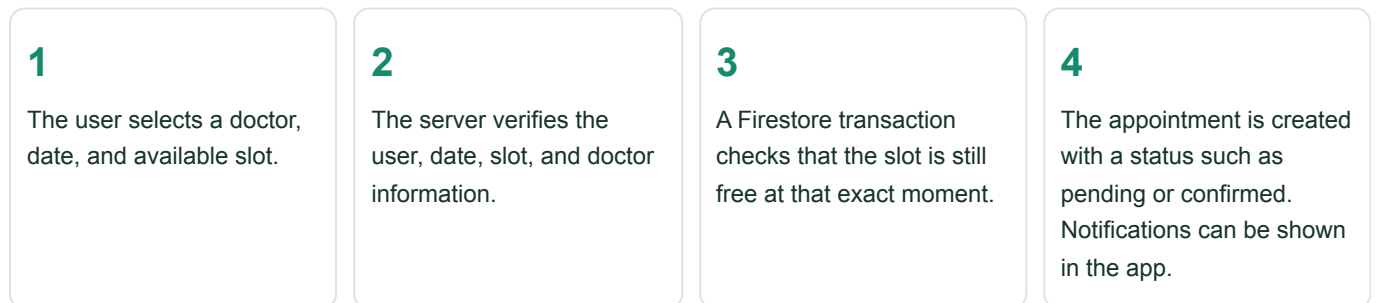
The visible interface does not connect directly to privileged services. It asks the Next.js server to perform protected operations. This keeps keys and sensitive decisions away from the browser.



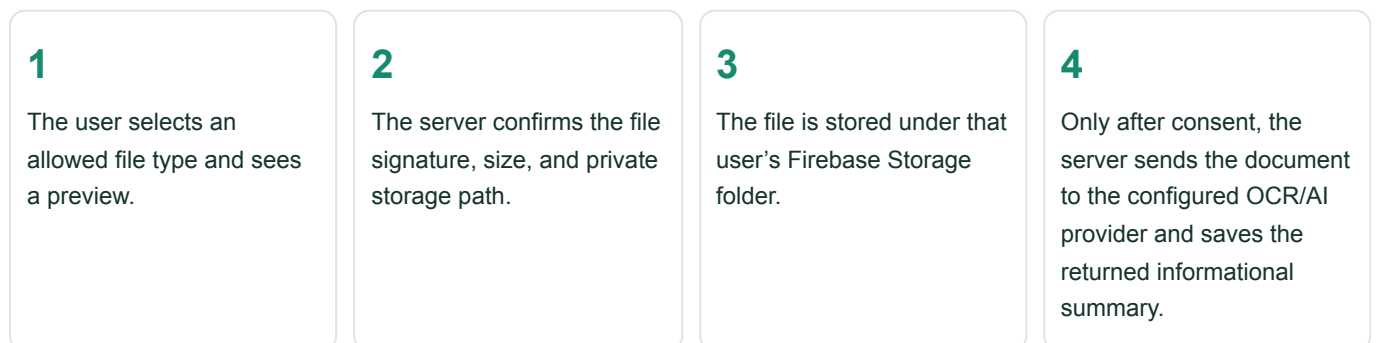
A simple example: saving water intake



A simple example: booking an appointment



A simple example: analysing a document



Why this approach is safer

- The Gemini key and Firebase service-account key exist only on the server.
- Browser pages use a short-lived Firebase sign-in token only to establish a protected server session.
- Server routes verify the session and role for each sensitive action.
- Database rules provide another layer of protection if a request reaches Firebase directly.

6. Data structure in plain language

Firestore stores information as named collections. A user's sensitive health information is held beneath their own account instead of in one shared public table.

Location	What is stored	Who can access it
<code>users/{userId}</code>	Name, email, role, contact preferences, and basic profile details.	The account owner and trusted server processes. Administrators use controlled management views.
<code>users/{userId}/healthReports</code>	Report metadata, optional analysis summary, and the list of doctors selected by the user.	The owner. A doctor only after a valid appointment relationship and explicit report sharing.
<code>users/{userId}/prescriptions</code>	Confirmed extraction results from the prescription reader.	The owner; access is not broadly shared by default.
<code>waterLogs</code> , <code>fitnessLogs</code> , <code>bmiRecords</code>	Daily wellbeing history that powers the health charts.	The owner.
<code>nutritionPlans</code> , <code>conversations</code> , <code>notifications</code>	Generated plans, AI chat messages, and in-app updates.	The owner.
<code>doctors/{doctorId}</code> and <code>availability</code>	Doctor profile data and available appointment slots.	Public doctor details are available for booking only after an admin/owner publishes a verified profile; the doctor manages their own schedule.
<code>appointments</code>	Patient, doctor, time, reason, and status.	The connected patient, connected doctor, and administrator according to server and Firebase rules.
<code>healthTips</code> , <code>emergencyContacts</code>	Administrator-managed education and emergency information.	Published/active entries are visible to signed-in users.
<code>feedback</code>	Feedback form submissions and their status.	The submitting user for their own action and administrators for management.

Where uploaded files go

File type	Storage location
Profile image	users/{userId}/profile/
Prescription file	users/{userId}/prescriptions/
Health report file	users/{userId}/health-reports/
Doctor-created prescription file	doctors/{doctorId}/prescriptions/

7. Security, privacy, and AI safety

Account protection

- Firebase Authentication handles passwords and login. Passwords are never stored in Firestore.
- Each account has a trusted server-assigned role: `user` , `doctor` , or `admin` . The browser cannot choose its own role.
- After sign-in, the app exchanges the Firebase token for an `httpOnly` session cookie. Browser JavaScript cannot read this cookie.
- Protected pages redirect unauthorised visitors to sign-in. Protected API routes repeat the role check on the server.

Data protection

- Firestore and Storage rules are included in the project: `firestore.rules` and `storage.rules` .
- Upload routes allow only PDF/JPG/JPEG/PNG, validate the actual file signature, limit uploads to 8 MB, and restrict files to private owner paths.
- Cross-origin state-changing requests are rejected using the configured application URL.
- Firebase App Check can be enabled after its reCAPTCHA configuration is complete, adding another layer against automated abuse.

AI safety rules built into the product

- Every AI experience states that it offers informational guidance and does not replace a clinician.
- The system prompt tells the AI not to claim it is a doctor, prescribe medicine, change a dosage, or invent missing findings.
- Emergency symptoms are directed toward immediate professional or emergency care.
- Report explanations are limited to text extracted from the uploaded document. If a reference range is missing, the app says so.
- Users must consent before a private document is sent to the configured AI service.

Before a public healthcare launch: deploy the Firebase rules, enable App Check, rotate any service-account key that has been shared, configure a production HTTPS domain, and have qualified legal, privacy, and healthcare professionals review local obligations.

8. Project folder structure

The following is a simplified map of the source code. It helps a new developer find the main areas without needing to know every file.

app/	Pages and secure API routes
(auth)/	Login, registration, password reset
(user)/	Patient pages: dashboard and health tools
doctor/	Doctor portal pages
admin/	Administrator portal pages
api/	Server-only actions for data, sessions, AI, uploads
components/	Reusable visual building blocks
dashboard/	Dashboard cards and charts
health/	Trackers, AI tools, role portals, staff views
reports/	Report and prescription upload views
layout/	Sidebar, mobile navigation, notifications
ui/	Buttons, inputs, dialogs, tables, empty states
lib/	Shared application logic
firebase/	Browser and server Firebase configuration
auth/	Session, role, and request-origin checks
firestore/	Data access helpers and access rules
ai/	Gemini prompts, parsing, response safety
ocr/	Replaceable OCR provider interface
storage/	Private upload paths and validation
validation/	Form and API validation rules
scripts/	Safe maintenance and sample-data tools
public/	Logo, illustrations, service worker
firebase.rules	Database security rules
storage.rules	File-storage security rules
README.md	Technical setup instructions

Reusable building blocks

The interface avoids rebuilding the same controls for every screen. Shared components include buttons, inputs, form fields, dialogs, cards, badges, tables, avatars, tabs, skeleton loading states, empty states, page headers, metric cards, file upload controls, AI response displays, doctor cards, and appointment cards.

9. Sample data and current setup

Sample records were added so the dashboards and staff areas can be explored without entering personal health information. These entries are marked **sample** and do not claim to be medical facts.

Seeded area	Sample content available
Patient accounts	Three supplied patient accounts with profile, health profile, seven-day water/fitness history, BMI history, report workspace record, prescription workspace record, nutrition plan, conversation, notification, and appointment.
Doctor accounts	Two supplied doctor accounts with availability slots and linked sample appointments. Their public profiles remain inactive until real professional details are reviewed and published.
Admin account	One supplied administrator account with data visible across appointment, feedback, health-tip, emergency-contact, user, doctor, and analytics pages.
Shared content	Published sample health tips, emergency contacts marked for local verification, feedback, and appointment records.

What needs to be configured for every feature to be live

1. **Cloud Storage:** In Firebase Console, open *Build* → *Storage* and create the bucket. Then deploy the included Storage rules. This enables file upload and retrieval.
2. **Gemini:** Create a Gemini key in Google AI Studio and add it to the server environment as `GEMINI_API_KEY` . Do not put it in browser code or share it in chat.
3. **Firebase rules:** Use Firebase CLI to deploy `firestore.rules` and `storage.rules` to the correct project.
4. **Production hosting:** Deploy to Firebase App Hosting, Vercel, or another Node-compatible platform. Add all environment variables there and set `APP_URL` to the real HTTPS domain.
5. **Doctor publication:** Add verified qualifications, specialisation, availability, fee, and profile description before making a doctor visible to patients.

Useful words explained

Word	Meaning
Frontend	The screens, buttons, and forms a person uses in the browser.
Backend	The protected server actions that check requests and safely connect to services.
Database	The organised cloud store for information such as profiles, appointments, and logs.
API	A controlled doorway used by the frontend to ask the backend to perform an action.
Role	The permission level of a user: patient, doctor, or administrator.
OCR	Technology that reads printed or handwritten-looking text from an image or document.

UyirNadi is designed as a healthcare-management MVP. For production clinical use, follow local laws, privacy requirements, and qualified professional review processes.